**De La Salle University - Dasmariñas**

**COLLEGE OF INFORMATION AND COMPUTER STUDIES**

**INFORMATION TECHNOLOGY DEPARTMENT**

# COURSE SYLLABUS

**First Semester, SY 2024-2025**

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|---|---|---|
| **COURSE CODE** | **:** | **S-ITPC312L** |
| **COURSE TITLE** | **:** | **Cisco Networking Semester 1** |
| **COURSE TYPE** | **:** | **Laboratory** |
| **COURSE CREDIT** | **:** | **One (1) unit** |
| **LEARNING MODALITY** | **:** | **Fully-onsite Learning Model** |
| **PRE-REQUISITES** | **:** | **S-ITCS214/LA** |
| **CO-REQUISITES** | **:** | **S-ITNT213LC** |
| **PROFESSOR** | **:** | **Maria Gloria Del Rosario** |
| **CONSULTATION HOURS** | **:** | **Thursday 7:00-12:00 online via MS Teams** |
| **CONTACT INFORMATON:** | | **madelrosario@dlsud.edu.ph (046) 4811900 local 3134** |

## VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

## MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

## COURSE DESCRIPTION:

Introduction to Networks (ITN) covers the architecture, structure, functions and components of the Internet and other computer networks. Students achieve a basic understanding of how networks operate and how to build simple local area networks (LAN), perform basic configurations for routers and switches, and implement Internet Protocol (IP).

## COURSE LEARNING OUTCOME

By the end of this course, students can:

CLO1. configure switches and end devices to provide access to local and remote network resources;
CLO2. explain how physical and data link layer protocols support the operation of Ethernet in a switched network;
CLO3. configure routers to enable end-to-end connectivity between remote devices;
CLO4. create IPv4 and IPv6 addressing schemes and verify network connectivity between devices;
CLO5. explain how the upper layers of the OSI model support network applications;
CLO6. use security best practices to configure a small network; and
CLO7. troubleshoot connectivity in a small network.

## LEARNING PLAN:

**Learning Outcomes for Midterm Period**

| **Course Learning Outcomes** | **Topic Learning Outcomes** |
|---|---|
| **Midterm Period** | |
| CLO1. Configure switches and end devices to provide access to local and remote network resources. | TLO1. Explain the advances in modern network technologies. |
| CLO2. Explain how physical and data link layer protocols support the operation of Ethernet in a switched network. | TLO2. Explain how physical layer protocols, services, and network media support communications across data networks. |
| CLO3. Configure routers to enable end-to-end connectivity between remote devices. | TLO3. Explain how routers use network layer protocols and services to enable end-to-end connectivity. |

| **Module Title and Learning Outcome(s)** | **Week** | **Teaching-Learning Activities / Assessment Strategies** | **Technology Enabler** | **Allotted Hours** |
|---|---|---|---|---|
| Module 0<br>CLO1<br>TLO1 | 1<br>Aug 19-24 | Gospel Reading, Reflection and Prayer<br>Presentation on the Fully online learning model LMS:<br>1. Schoolbook enlistment<br>2. MS Teams channel for the course<br>Netacad Course Experience – Enrollment, Interface, Submission, Getting Used to the Cisco Networking Academy LMS<br>3. ICT213 Laboratory Orientation. Learn Packet Tracer GUI<br><br>Navigate the IOS<br>Configure Initial Switch Settings<br>Implement Basic Connectivity | MS Teams<br>Schoolbook<br>NetAcad | 3 |
| Module 1<br>CLO1<br>TLO1 | 2<br>Aug 26<br>-31 | Module 1: Networking Today<br>Gospel Reading, Reflection and Prayer<br>Lab Setup: Basic Switch and End Device Configuration<br>Implement initial settings including passwords, IP addressing, and default gateway parameters on a network switch and end devices.<br><br>Enabling Activity 1: PT Network Representation | MS Teams<br>Schoolbook<br>NetAcad | 3 |
| Module 1<br>CLO1<br>TLO1 | 3<br>Sep 2-7 | Gospel Reading, Reflection and Prayer<br>Module 1: Networking Today<br>Objective: Explain the advances in modern network technologies<br>Test the Interface Assignment<br>Test End-to-End Connectivity<br>Enabling Activity 2: Basic Switch and End Device Conf | MS Teams<br>Schoolbook<br>NetAcad | 3 |

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| Module 2<br>CLO2<br>TLO2 | 4<br>Sep 9-14 | Gospel Reading, Reflection and Prayer<br>Module 2: Protocols and Models<br>Objective: Explain how network protocols enable devices to access local and remote network resources<br>Download Wireshark<br>Enabling Activity 3: Use Wireshark to View Network Traffic | MS Teams<br>Schoolbook<br>NetAcad | 3 |
| Module 3<br>CLO2<br>TLO2 | 5<br>Sep 16-21 | Gospel Reading, Reflection and Prayer<br>Module 3: Number Systems<br>Objective: Calculate numbers between decimal, binary and hexadecimal systems<br>Investigate the TCP-IP and OSI Models in Action<br>Enabling Activity 4: Connect to the Physical Layer | MS Teams<br>Schoolbook<br>NetAcad | 3 |
| Module 4<br>CLO2<br>TLO2 | 6<br>Sep 23-28 | Gospel Reading, Reflection and Prayer<br>Module 4: Data Link Layer<br>Objective: Explain how media access control in the data link layer supports communication across networks Enabling Activity 5: View Wired and Wireless NIC Info | MS Teams<br>Schoolbook<br>NetAcad | 3 |
| Module 5<br>CLO2<br>TLO2 | 7<br>Sep 30 – Oct 5 | Gospel Reading, Reflection and Prayer<br>Module 5: Network Layer<br>Objective: Explain how routers use network layer protocols and services to enable endto-end connectivity<br>Enabling Activity 6: Use Wireshark to Examine Ethernet Frames | MS Teams<br>Schoolbook<br>NetAcad | 3 |
| Module 1,2,3,4,5 | 8<br>Oct 7-12 | Summative Assessment based on Module 1-5 | MS Teams<br>Schoolbook<br>NetAcad | 3 |
| | | | TOTAL | 24 |

**LEARNING PLAN FOR FINALS:**

| **Course Learning Outcomes** | **Topic Learning Outcomes** |
|---|---|
| CLO4. Create IPv4 and IPv6 addressing schemes and verify network connectivity between devices. | TLO4. Implement a VLSM addressing scheme.<br>TLO5. Implement a subnetted IPv6 addressing scheme. |

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| CLO5. Explain how the upper layers of the OSI model support network applications. | TLO6. Compare the operations of transport layer protocols in supporting end-to-end communication.<br>TLO7. Explain the operation of application layer protocols in providing support to enduser applications. |
| CLO6. Use security best practices to configure a small network. | TLO8. Configure switches and routers with device hardening features to enhance security. |
| CLO7. Troubleshoot connectivity in a small network. | TLO9. Implement a network design for a small network to include a router, a switch, and end devices. |

| **Module Title and Learning Outcome(s)** | **Week** | **Teaching-Learning Activities / Assessment Strategies** | **Technology Enabler** | **Allotted Hours** |
|---|---|---|---|---|
| Module 6<br>CLO4<br>TLO4 | 10<br>Oct 21-26 | Gospel Reading, Reflection and Prayer<br>Module 6: IPv4 Addressing<br>Objective: Calculate an IPv4 subnetting scheme to efficiently segment a network<br>Subnetting Scenario<br>Subnet an IPv4 Network<br>Enabling Activity 1: Design and Implement a VLSM Addressing Scheme | | 3 |
| Module 6<br>CLO4<br>TLO4 | 11<br>Oct 28-Nov2 | Independent Learning Week | | 3 |
| Module 6<br>CLO4<br>TLO4 | 12<br>Nov 4-9 | Gospel Reading, Reflection and Prayer<br>Module 6: IPv6 Addressing<br>Objective: Explain the need for IPv6<br>Subnetting Scenario<br>Subnet an IPv6 Network<br>Enabling Activity 2: Verify IPv4 and IPv6 Addressing | MS Teams<br>Schoolbook<br>NetAcad | 3 |
| Module 6<br>CLO4<br>TLO4 | 13<br>Nov 11-16 | Gospel Reading, Reflection and Prayer<br>Module 6: IPv6 Addressing<br>Objective: Calculate an IPv6 subnetting scheme to efficiently segment a network<br>Enabling Activity 3: Configure IPv6 Addressing | MS Teams<br>Schoolbook<br>NetAcad | 3 |

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|---|---|---|---|---|
| Module 7<br>CLO4<br>TLO5 | 14<br>Nov 18-23 | Gospel Reading, Reflection and Prayer<br>Module 7: IPv6 Addressing<br>Objective: Explain the need for IPv6 addressing<br>IPv6 Neighbor Discovery<br>Implement a Subnetted IPv6 Addressing Scheme<br>Enabling Activity 4: Implement a Subnetted IPv6 Addressing Scheme | MS Teams<br>Schoolbook<br>NetAcad | 3 |
| Module 8<br>CLO5<br>TLO6<br>TLO7 | 15<br>Nov 25-30 | Gospel Reading, Reflection and Prayer<br>Module 8: ICMP<br>Objective: Use various tools to test network connectivity<br>Use Ping and Traceroute to Test Network Connectivity<br>Enabling Activity 5: Using ICMP to Test and Correct Network Connectivity | MS Teams<br>Schoolbook<br>NetAcad | 3 |
| Module 9<br>CLO5<br>TLO6<br>TLO7 | 16<br>Dec 2-7 | Gospel Reading, Reflection and Prayer<br>Module 9: Transport Layer<br>Objective: Compare the operations of transport layer protocols in supporting endto-end communication<br>PT on TCP and UDP Communications<br>Enabling Activity 6: Configure Secure Password with SSH | MS Teams<br>Schoolbook<br>NetAcad | 3 |
| Module 6,7,8,9 | 17<br>Dec 9-14 | Summative Assessment: Skills-based Activity | MS Teams<br>Schoolbook<br>NetAcad | 3 |
| | | | **TOTAL** | 24 |

**FINAL PRODUCT/PERFORMANCE:**

All students taking this course must accomplish the following requirements listed below in order to gain the certificate of completion for Introduction to Networks.

- Complete all online chapter exams (Chapter 1-11, PT Practice Skills Assessment) with a minimum grade of 80 for all chapter exams.
- Complete the online final exam with a minimum grade of 80.
- Accomplish the teacher-made test and course feedback online.
- Pass the Hands-on Skills Lab.
- Submit the required PT activities completed to 100%.
- Demonstrate mastery on the interactive activity on key terms and concepts.

**GRADING SYSTEM:**

| | | **Midterm** |
|---|---|---|
| Enabling Assessments | - | 40% |
| Summative Assessments | - | 60% |
| Total | | 100% |

| **Finals** | | |
|---|---|---|
| Enabling Assessments | - | 40% |
| Summative Assessments | - | 60% |
| Total | | 100% |

Overall Final Grade = [(Midterm grade) + (Final term grade)] / 2

**COURSE POLICIES AND REQUIREMENTS:**

1. This course follows the Onsite Learning Mode. All activities will be done in the laboratory classroom and field.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar.

3. Laboratory Policies
   a. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
   b. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones also cannot be used as calculator during examination.
   d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

4. Communication and Feedback
   a. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
      i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
      ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
      iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
   b. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

5. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.

c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

6. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

7. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.

**REFERENCES:**


| Call number or e-provider | Reference Material |
|---|---|
| E-Book 2332775 | Davies G. 2019. Networking Fundamentals. Packt Publishing Ltd. |
| E-Book 3038316 | VandenBrink R. 2021. Linux for Networking Professionals. Packt Publishing Ltd. |
| E-Book 2022991 | Abhishek Ratan, Eric Chou, Pradeeban Kathiravelu, & Dr. M. O. Faruque Sarker. (2019). Python Network Programming : Conquer All Your Networking Challenges with the Powerful Python Language. Packt Publishing. |
| E-Book edspgr.11757827 | Cisco Technology, I. (San J. C. (2023). Network security from host and network impersonation. |
| QA 76.73.P98 .Or82 2019 | Sarker, Dr. M. 2019. Learning Python networking : a complete guide to build and deploy strong networking capabilities using Python 3.7 and Ansible. Second edition. UK : Packt Publishing, Ltd. |
| IEEE Xplore | Chadha, D. 2019 Software-Defined Optical Networks, IEEE. Wiley-IEEE Press |
| IEEE Xplore | Panek,C. 2020. Networking Fundamentals. Wiley. Wiley Data and Cybersecurity. |
| Website | Login \| Skills for All with Cisco. 2024. Netacadcom. [accessed 2024 Aug 5]. https://lms.netacad.com/mod/lti/view.php?id=82254473. |

**Prepared:** **INFORMATION TECHNOLOGY DEPARTMENT**
1st Semester 2024-2025

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